CHAPTER 17

DRAINAGE SYSTEM

CHAPTER ESSENCE AT A GLANCE:
Examine geometric drainage patterns (dendritic, radial, trellis, centripetal), Himalayan river systems (Indus, Ganga, Brahmaputra) versus Peninsular drainage (Godavari, Krishna, Narmada, Tapi), water divides, and river discharge regimes.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Drainage Patterns and Structural Geometries:** Dendritic pattern develops on uniform lithology resembling tree branches; Trellis pattern develops in folded alternating hard and soft rocks with right-angle tributaries; Radial pattern originates from an elevated dome or central peak (Amarkantak).

• **Himalayan Antecedent Drainage Systems:** Perennial river systems fed by snowmelt and monsoon rain; antecedent drainage carved deep V-shaped gorges through rising mountain ranges; Indus (emptying into Arabian Sea), Ganga, and Brahmaputra systems.

• **Peninsular Drainage and Graded Profiles:** Old, graded river valleys flowing down gentle slopes; strictly dependent on seasonal monsoon precipitation; major east-flowing rivers (Godavari, Krishna, Kaveri, Mahanadi) building deltas, and west-flowing fault rift rivers (Narmada, Tapi).

• **River Regimes and Hydrological Discharge:** Discharge regime reflects seasonal streamflow volume variation (Q); Himalayan rivers experience two high-water peaks (summer snowmelt and monsoon), while Peninsular rivers show extreme monsoon flood peaks and summer trickles.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$Q = A \times v \rightarrow \text{Streamflow Discharge } (\text{m}^3\ \text{s}^{-1}) = \text{Area } (\text{m}^2) \times \text{Velocity } (\text{m}\ \text{s}^{-1})$$

Hydrological Continuity Law of River Discharge

$$D_d = \frac{\sum L_{\text{channels}}}{A_{\text{basin}}} \rightarrow \text{Drainage Basin Channel Density Metric}$$

Horton Drainage Density Structural Formula

$$\text{Continental Water Divide: Arabian Sea } (23\%\ \text{Area}) \longleftrightarrow \text{Bay of Bengal } (77\%\ \text{Area})$$

Primary Hydrological Catchment Partition of India

$$\frac{Q_{\text{Monsoon}}}{Q_{\text{Lean}}} \rightarrow \text{Peninsular Seasonal Discharge Ratio} \gg \text{Himalayan Ratio}$$

Regime Volatility Ratio of Rain-Fed Peninsular Streams

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Narmada and Tapi flow westwards into the Arabian Sea not because of regional relief slope, but because they flow through structural rift valleys (tectonic fault grabens) created by crustal subsidence.

• **Alert:** East-flowing rivers (Godavari, Krishna, Mahanadi) build expansive coastal deltas at their mouths, whereas west-flowing rivers (Narmada, Tapi) form estuaries due to steep gradients and hard rocky beds.

• **Alert:** The Indus Water Treaty of 1960 allocates the exclusive waters of three western rivers (Indus, Jhelum, Chenab) primarily to Pakistan, while three eastern rivers (Ravi, Beas, Sutlej) are allocated to India.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Dendritic and Radial drainage patterns with examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Dendritic Pattern:** A branching drainage pattern that resembles the branches of a tree, developing on uniform terrain underlain by rocks of uniform resistance (e.g. rivers of Northern Plains like Ganga). **[1 Mark]**

• **Radial Pattern:** Develops when streams flow outward in all directions from a central elevated dome or volcanic peak (e.g. rivers originating from Amarkantak mountain like Narmada and Son). **[1 Mark]**

2. Why do the rivers Narmada and Tapi flow westwards and form estuaries instead of deltas? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Westward Flow:** They flow westwards into the Arabian Sea because they occupy linear structural rift valleys (fault troughs) created during Himalayan tectonic collisions. **[1 Mark]**

• **Estuary Formation:** They flow through hard basaltic and rocky terrains, carrying very low sediment loads, and have steep channel gradients preventing delta building. **[1 Mark]**

3. Compare the Himalayan rivers with the Peninsular rivers on the basis of origin, nature of flow, and stage of evolution. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Origin:** Himalayan rivers originate from snow-covered glaciers and perennial springs in the high Himalayas; Peninsular rivers originate from plateaus and Western Ghats. **[1 Mark]**

• **Nature of Flow:** Himalayan rivers are perennial (fed by both snowmelt and rain); Peninsular rivers are seasonal (strictly dependent on monsoon rainfall). **[1 Mark]**

• **Stage of Evolution:** Himalayan rivers are young, vigorous, cutting deep V-shaped gorges; Peninsular rivers are old, mature, flowing through shallow graded valleys. **[1 Mark]**

4. What is a Water Divide? Give two examples of major water divides in the Indian subcontinent. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** An elevated ridge, mountain, or upland line separating two adjacent river drainage basins, causing water to flow into different river systems. **[1 Mark]**

• **Examples:** (1) The Delhi-Aravalli ridge separating the Indus drainage basin from the Ganga drainage basin, (2) The Western Ghats acting as the major water divide between Arabian Sea drainage and Bay of Bengal drainage. **[2 Marks]**

5. Describe the Ganga River System, mentioning its headstreams, major right-bank and left-bank tributaries, and its delta. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Headstreams:** Originates as Bhagirathi from Gangotri glacier; joins Alaknanda at Devprayag to form the river Ganga, which emerges into the northern plains at Haridwar. **[1 Mark]**

• **Right-Bank Tributaries:** Yamuna is the largest tributary originating from Yamunotri glacier and joining at Prayagraj; other peninsular tributaries include Son and Damodar. **[1 Mark]**

• **Left-Bank Tributaries:** Major tributaries include Ramganga, Gomati, Ghaghara, Gandak, Kosi, and Mahananda flowing southwards from the Himalayas. **[1 Mark]**

• **Delta Formation:** Splits into distributaries Bhagirathi-Hooghly and Padma in Bangladesh, joining Jamuna (Brahmaputra) and Meghna to form the world's largest Sundarbans Delta before entering Bay of Bengal. **[1 Mark]**

6. Explain the concept of 'River Regime'. Compare the hydrological regimes of the Ganga and the Narmada rivers. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **River Regime Concept:** The seasonal pattern of flow and fluctuations in discharge of water in a river channel over the course of a year. **[1 Mark]**

• **Ganga River Regime:** Perennial regime characterized by two distinct high discharge peaks: a minor peak in spring (April-May) caused by Himalayan snowmelt, and a massive primary flood peak in August-September caused by monsoon rains. **[1.5 Marks]**

• **Narmada River Regime:** Strictly rain-fed seasonal regime; maintains a very low discharge trickle during the dry winter and summer months (January to June), followed by an explosive peak discharge during August due to concentrated monsoon rainfall. **[1.5 Marks]**